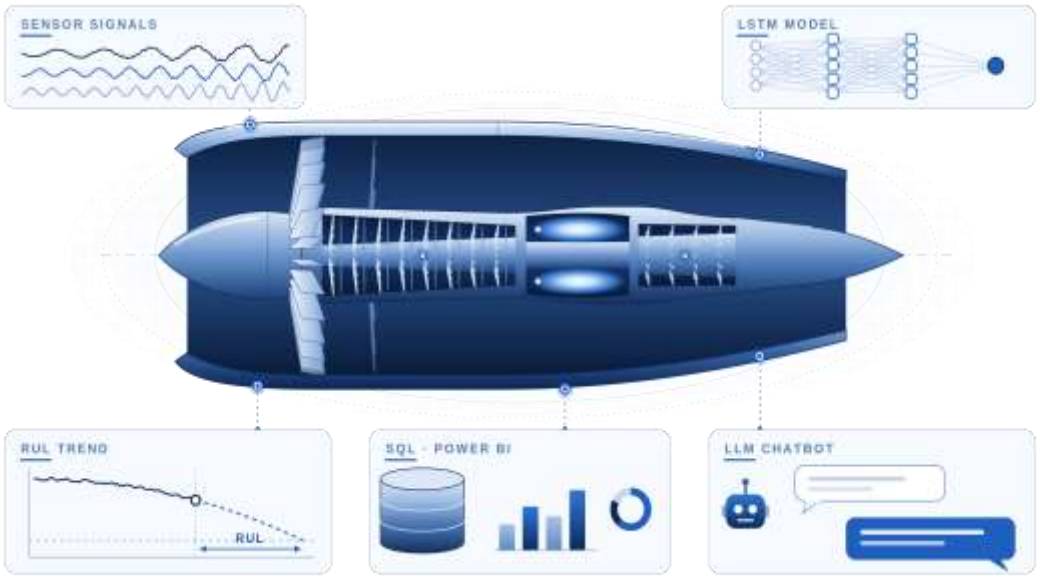
Predictive Maintenance for Aircraft Engines

NASA Turbofan Engine RUL Prediction



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Chapter 1
Project Introduction

This chapter introduces the project background, defines the problem being addressed, outlines the main project objectives, and describes the project scope and key development stages.

1.1 Project Background

Aircraft engines operate under demanding conditions, making regular maintenance and early fault detection essential for safe and reliable operation. Traditional maintenance strategies may not always provide an accurate estimate of an engine's remaining operational life.

Predictive Maintenance uses data analysis and machine learning techniques to monitor equipment conditions and estimate potential failures before they occur. In this project, the NASA Turbofan Jet Engine Dataset is used to analyze engine sensor readings and develop a system for Remaining Useful Life (RUL) prediction. The project combines data analysis, deep learning, database management, interactive visualization, and Large Language Models (LLMs) to support engine health monitoring and maintenance planning.

1.2 Problem Statement

Aircraft engines generate large amounts of sensor data during operation. However, analyzing these readings to estimate the Remaining Useful Life (RUL) of an engine is challenging due to changing operating conditions and complex degradation patterns. Without reliable RUL estimates, planning maintenance activities and identifying engines that require attention can be difficult.

Therefore, this project aims to develop a data-driven system that analyzes engine sensor readings and predicts the estimated remaining useful life of each engine. The system also integrates database management, interactive dashboards, and a Large Language Model (LLM)-based chatbot to make prediction results more accessible and easier to understand.

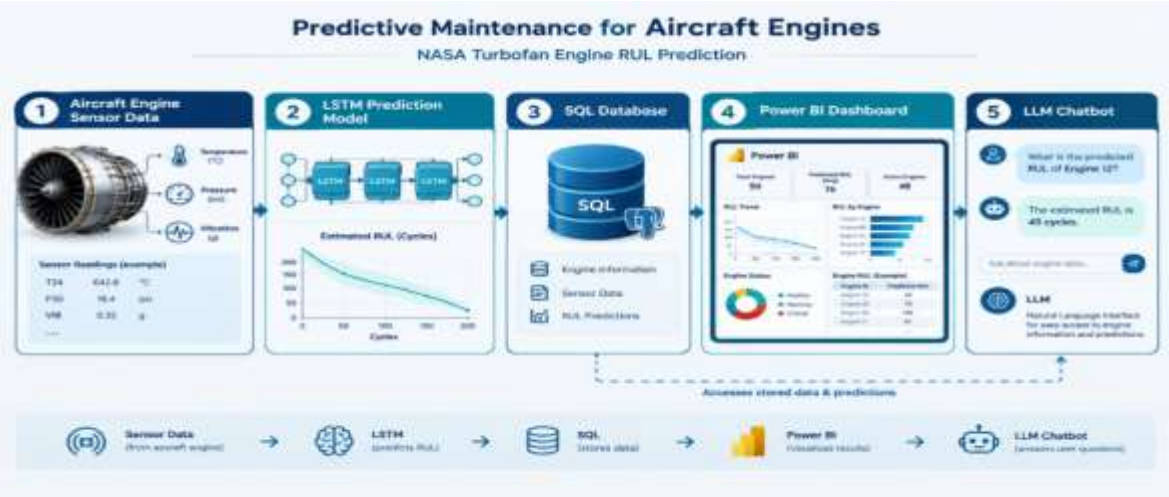
1.3 Project Objectives

The main objectives of this project are to:

1. Analyze and visualize the NASA Turbofan Jet Engine Dataset to understand sensor readings and engine degradation patterns.
2. Develop an LSTM-based deep learning model to predict the Remaining Useful Life (RUL) of aircraft engines.
3. Design and implement an SQL database to store and manage engine data and RUL predictions.
4. Develop interactive Power BI dashboards to visualize engine performance and key performance indicators (KPIs).
5. Integrate a Large Language Model (LLM)-based chatbot to allow users to query engine information and retrieve RUL predictions through natural language.
6. Integrate and test the system components to provide a unified solution for engine health monitoring and predictive maintenance.

1.4 Project Scope

The following figure illustrates the main stages of the proposed project, from data preparation and analysis to RUL prediction, database integration, dashboard development, and chatbot implementation, ending with system integration and testing.


Predictive Maintenance for Aircraft Engines
NASA Turbofan Engine RUL Prediction
1 Aircraft Engine Sensor Data
2 LSTM Prediction Model
3 SQL Database
4 Power BI Dashboard
5 LLM Chatbot
SQL
LLM
Accesses stored data & predictions
Sensor Data
LSTM
SQL
Power BI
LLM Chatbot

Chapter 2
Dataset Understanding

This chapter provides an overview of the NASA Turbofan Jet Engine Dataset, including its structure, subsets, and operating conditions. It also explains the concept of Remaining Useful Life (RUL) and its role in the project's prediction task.

2.1 Dataset Overview and Structure

The NASA Turbofan Jet Engine Dataset is a publicly available dataset used for research in prognostics and predictive maintenance. It contains simulated engine run-to-failure data, including operational settings and sensor measurements recorded over successive operating cycles.

The dataset consists of four subsets: FD001, FD002, FD003, and FD004, which differ in operating conditions and fault modes. Each data record represents an engine's operating state during a specific cycle, allowing the analysis of changes in sensor readings over time and supporting the development of Remaining Useful Life (RUL) prediction models.

The dataset includes engine identifiers, cycle numbers, operational settings, and multiple sensor measurements. These variables provide the information needed to study engine degradation patterns and develop a data-driven approach for estimating the remaining operational life of each engine.

The dataset used in this project was obtained from Kaggle. The original dataset was provided by NASA's Prognostics Center of Excellence (PCoE).

Dataset Source: https://www.kaggle.com/datasets/behrad3d/nasa-cmaps

ID	Cycle	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	S16	S17	S18	S19	S20	S21
1	1	518.67	641.82	1589.7	1400.6	14.62	21.61	554.36	2388.06	9046.19	1.3	47.47	521.66	2388.02	8138.62	8.4195	0.03	392	2388	100	39.06	23.419
1	2	518.67	642.15	1591.82	1403.14	14.62	21.61	553.75	2388.04	9044.07	1.3	47.49	522.28	2388.07	8131.49	8.4318	0.03	392	2388	100	39	23.4236
1	3	518.67	642.35	1587.99	1404.2	14.62	21.61	554.26	2388.08	9052.94	1.3	47.27	522.42	2388.03	8133.23	8.4178	0.03	390	2388	100	38.95	23.3442
1	4	518.67	642.35	1582.79	1401.87	14.62	21.61	554.45	2388.11	9049.48	1.3	47.13	522.86	2388.08	8133.83	8.3682	0.03	392	2388	100	38.88	23.3739
1	5	518.67	642.37	1582.85	1406.22	14.62	21.61	554	2388.06	9055.15	1.3	47.28	522.19	2388.04	8133.8	8.4294	0.03	393	2388	100	38.9	23.4044
1	6	518.67	642.1	1584.47	1398.37	14.62	21.61	554.67	2388.02	9049.68	1.3	47.16	521.68	2388.03	8132.85	8.4108	0.03	391	2388	100	38.98	23.3669
1	7	518.67	642.48	1592.32	1397.77	14.62	21.61	554.34	2388.02	9059.13	1.3	47.36	522.32	2388.03	8132.32	8.3974	0.03	392	2388	100	39.1	23.3774
1	8	518.67	642.56	1582.96	1400.97	14.62	21.61	553.85	2388	9040.8	1.3	47.24	522.47	2388.03	8131.07	8.4076	0.03	391	2388	100	38.97	23.3106
1	9	518.67	642.12	1590.98	1394.8	14.62	21.61	553.69	2388.05	9046.46	1.3	47.29	521.79	2388.05	8125.69	8.3726	0.03	392	2388	100	39.05	23.4066
1	10	518.67	641.71	1591.24	1400.46	14.62	21.61	553.59	2388.05	9051.7	1.3	47.03	521.79	2388.06	8129.38	8.4286	0.03	393	2388	100	38.95	23.4694
1	11	518.67	642.28	1581.75	1400.64	14.62	21.61	554.54	2388.05	9049.61	1.3	47.15	521.4	2388.01	8140.58	8.434	0.03	392	2388	100	38.94	23.4767
1	12	518.67	642.06	1583.41	1400.15	14.62	21.61	554.52	2388.09	9049.37	1.3	47.18	521.8	2388.02	8134.25	8.3938	0.03	391	2388	100	39.06	23.366
1	13	518.67	643.07	1582.19	1400.83	14.62	21.61	553.44	2388.12	9046.82	1.3	47.38	521.85	2388.08	8128.1	8.4152	0.03	393	2388	100	38.93	23.2757
1	14	518.67	642.35	1592.95	1399.16	14.62	21.61	554.48	2388.09	9047.37	1.3	47.44	521.67	2388	8134.43	8.3964	0.03	393	2388	100	39.18	23.3826
1	15	518.67	642.43	1583.82	1402.13	14.62	21.61	553.64	2388.11	9052.22	1.3	47.3	522.5	2388.08	8127.56	8.4199	0.03	391	2388	100	38.99	23.35
1	16	518.67	642.13	1587.98	1404.5	14.62	21.61	553.94	2388.05	9049.34	1.3	47.24	521.49	2388.07	8136.11	8.3936	0.03	392	2388	100	38.97	23.455
1	17	518.67	642.58	1584.96	1399.95	14.62	21.61	553.8	2388.06	9054.92	1.3	47.12	521.89	2388.04	8137.27	8.4542	0.03	392	2388	100	38.81	23.3319
1	18	518.67	642.62	1591.04	1396.12	14.62	21.61	554.2	2388.05	9049.55	1.3	47.21	521.76	2388.07	8132.73	8.4026	0.03	392	2388	100	38.89	23.3987
1	19	518.67	641.79	1587.56	1400.35	14.62	21.61	554.18	2388.04	9053.99	1.3	47.4	521.89	2388.03	8129.13	8.4321	0.03	391	2388	100	38.8	23.3464

Partial View of dataset sample

2.2 Dataset Subsets and Operating Conditions

The NASA Turbofan Jet Engine Dataset is divided into four subsets: FD001, FD002, FD003, and FD004. These subsets represent different combinations of operating conditions and engine fault modes, providing varying levels of complexity for engine degradation analysis and RUL prediction.

FD001 and FD002 represent a single fault mode, High-Pressure Compressor (HPC) degradation, while FD003 and FD004 include two fault modes: HPC degradation and fan degradation. The subsets also differ in operating conditions, as summarized in Table 2.1.

Characteristics of the NASA Turbofan Dataset Subsets:

Subset	Operating Conditions	Fault Modes
FD001	1	HPC
FD002	6	HPC
FD003	1	HPC and Fan
FD004	6	HPC and Fan

2.3 Remaining Useful Life (RUL)

Remaining Useful Life (RUL) refers to the estimated number of operational cycles an engine can complete before reaching its end-of-life condition. In the NASA Turbofan Jet Engine Dataset, each engine is monitored over successive cycles, allowing its degradation to be analyzed over time.

RUL is the target variable in this project, as the LSTM model will use historical sensor measurements to estimate the remaining cycles for each engine. These predictions can then be stored in the SQL database and presented through the Power BI dashboard and the LLM-based chatbot.

Chapter 3
AI Model Development and Evaluation

This chapter describes the development of the AI models used to predict the Remaining Useful Life (RUL) of aircraft engines using the NASA C-MAPSS dataset. It presents the data preparation process, model architectures, training procedures, and evaluation methods for each dataset subset, followed by a comparison of the obtained results.

3.1 Introduction to the AI Approach

The project uses Artificial Intelligence (AI) and Deep Learning techniques to predict the Remaining Useful Life (RUL) of aircraft engines. The prediction is based on sensor readings collected during engine operation.

Long Short-Term Memory (LSTM), a type of Recurrent Neural Network (RNN), is used to learn patterns from sequential sensor data and estimate the number of remaining cycles before an engine reaches the end of its useful life. The models are developed and evaluated using the four NASA C-MAPSS dataset subsets: FD001, FD002, FD003, and FD004.

3.2 FD001 Model Development and Evaluation

Project Notebook For FD001 data Set

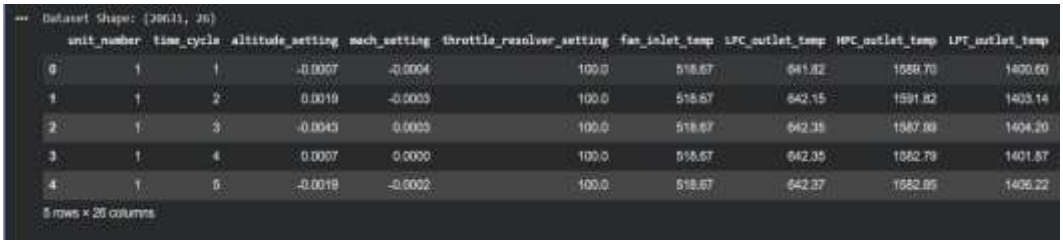
The complete implementation of the FD001 RUL prediction model is available in the following notebook:

Notebook Link:
https://colab.research.google.com/drive/1B3Ye6s9yWjMsU4mX6d2gMBOD7ImstDUw#scrollTo=78fb4c8e

3.2.1 Dataset Preparation and RUL Calculation

The FD001 subset of the NASA C-MAPSS dataset was used to develop a model for predicting the Remaining Useful Life (RUL) of aircraft engines. The dataset includes engine identifiers, operating cycles, three operational settings, and 21 sensor measurements. The training data contains engine records collected until failure, while the test data contains partial operating histories used to predict the remaining useful life of each engine.

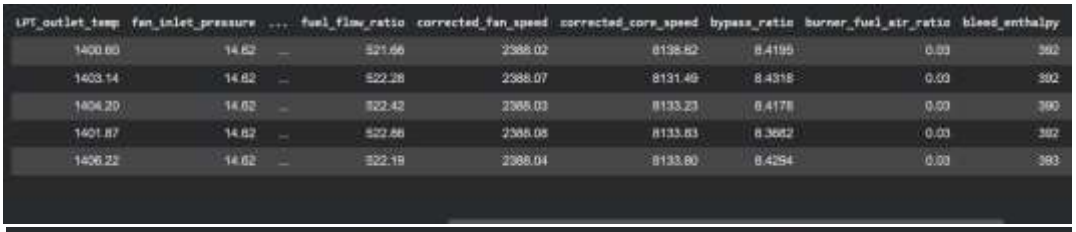
For the training data, the RUL was calculated by subtracting the current operating cycle from the maximum recorded cycle of the corresponding engine. This value represents the number of remaining operating cycles before the engine reaches failure.



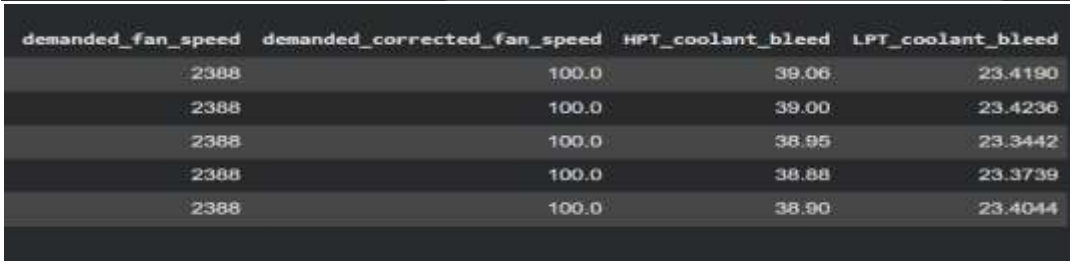
Dataset Shape: (20631, 26)

	unit_number	time_cycle	altitude_setting	mach_setting	throttle_resolver_setting	fan_inlet_temp	LPC_outlet_temp	HPC_outlet_temp	LPT_outlet_temp
0	1	1	-0.0007	-0.0004	100.0	518.67	641.82	1589.70	1400.60
1	1	2	0.0019	-0.0003	100.0	518.67	642.15	1591.82	1403.14
2	1	3	-0.0043	0.0003	100.0	518.67	642.35	1587.99	1404.20
3	1	4	0.0007	0.0000	100.0	518.67	642.35	1582.79	1401.87
4	1	5	-0.0019	-0.0002	100.0	518.67	642.37	1582.85	1406.22

5 rows × 26 columns



LPT_outlet_temp	fan_inlet_pressure	...	fuel_flow_ratio	corrected_fan_speed	corrected_core_speed	bypass_ratio	burner_fuel_air_ratio	bleed_enthalpy
1400.60	14.62	...	521.66	2388.02	8138.62	8.4195	0.03	392
1403.14	14.62	...	522.28	2388.07	8131.49	8.4318	0.03	392
1404.20	14.62	...	522.42	2388.03	8133.23	8.4178	0.03	390
1401.87	14.62	...	522.86	2388.08	8133.83	8.3682	0.03	392
1406.22	14.62	...	522.19	2388.04	8133.80	8.4294	0.03	393



demanded_fan_speed	demanded_corrected_fan_speed	HPT_coolant_bleed	LPT_coolant_bleed
2388	100.0	39.06	23.4190
2388	100.0	39.00	23.4236
2388	100.0	38.95	23.3442
2388	100.0	38.88	23.3739
2388	100.0	38.90	23.4044

Sample Records and Feature Columns of the FD001 Training Dataset.

```
# ...[] value RUL
def compute_rul(df):
  max_cycles = df.groupby("unit_number")["time_cycle"].max()
  merged = df.merge(max_cycles.rename("max_cycle"), on="unit_number")
  df["RUL"] = merged["max_cycle"] - merged["time_cycle"]
  return df

train_df = compute_rul(train_df)

print("Sample with Calculated RUL:")
train_df[["unit_number", "time_cycle", "RUL"]].head()
```

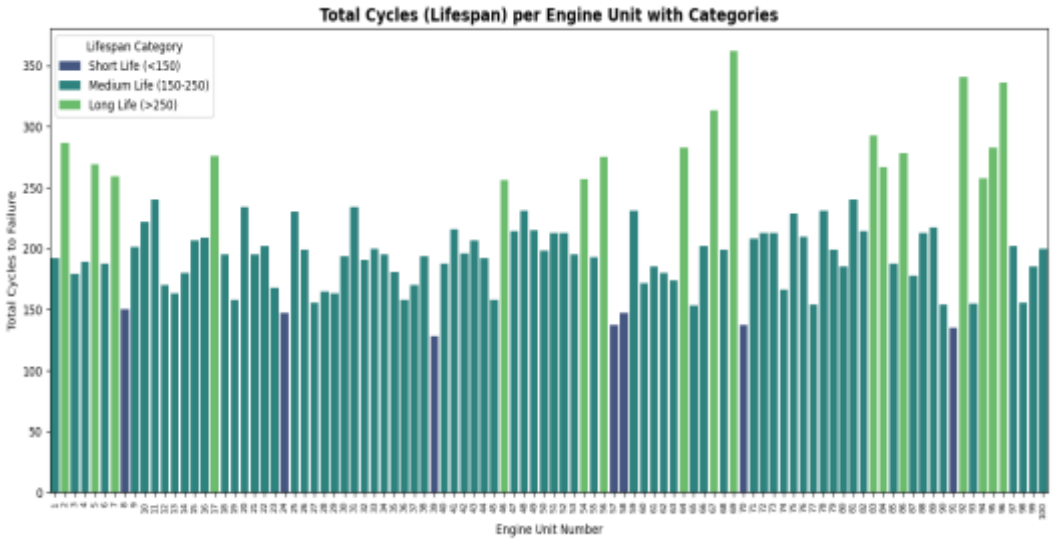
Sample with Calculated RUL:

	unit_number	time_cycle	RUL
0	1	1	191
1	1	2	190
2	1	3	189
3	1	4	188
4	1	5	187

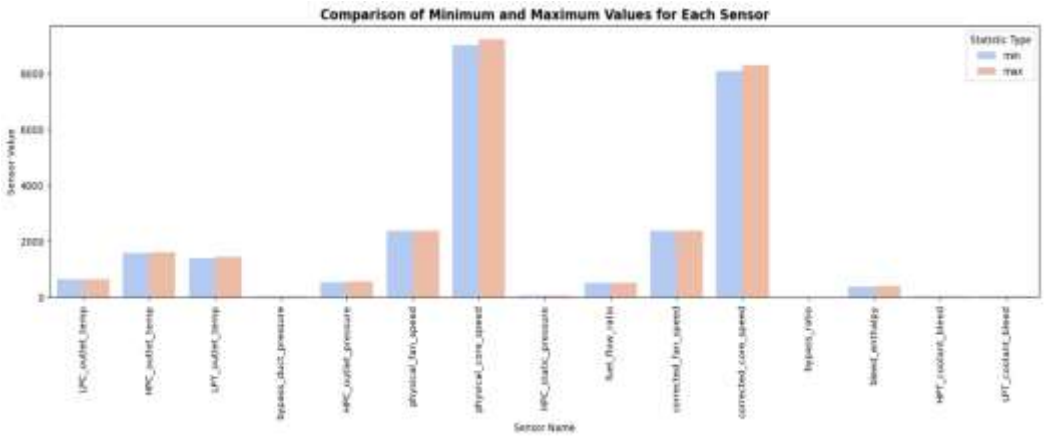
Remaining Useful Life (RUL) Calculation for the FD001 Training Data.

3.2.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the operating lifetimes of the engines and examine the sensor measurements in the FD001 dataset. The analysis focused on the number of operating cycles recorded for each engine and the variation in sensor values. These observations helped provide an initial understanding of the data before feature engineering and model development.



Total Operating Cycles per Engine in the FD001 Training Dataset.



Comparison of Minimum and Maximum Values for Each Sensor

3.2.3 Feature Engineering and Data Scaling

Feature engineering was performed to extract additional temporal information from the sensor measurements. For each engine, the rolling mean and rolling standard deviation were calculated using a window of 15 operating cycles. These features were generated separately for each engine to preserve its operating history. Missing values resulting from the rolling standard deviation calculation were replaced with zero.

The RUL target values were capped at 125 cycles. Min-Max scaling was then applied to the input features. The scaler was fitted using the training data and subsequently applied to the validation and test data to maintain a consistent feature scaling process.

```
# 2. إضافة المميزات المعتمدة: Rolling Mean + Rolling Std
window_size = 15
for df in [train_df, test_df]:
  for col in base_feature_cols:
    df[f"{col}_mean"] = (
        df.groupby("unit_number")[col]
        .transform(lambda x: x.rolling(window=window_size, min_periods=1).mean())
    )
    df[f"{col}_std"] = (
        df.groupby("unit_number")[col]
        .transform(lambda x: x.rolling(window=window_size, min_periods=1).std())
        .fillna(0)
    )
```

Rolling Mean and Rolling Standard Deviation Feature Engineering.

```
# 4. توحيد النطاق (Scaling)
scaler = MinMaxScaler()
train_data_scaled = train_data.copy()
val_data_scaled = val_data.copy()
test_data_scaled = test_df.copy()

train_data_scaled[feature_cols] = scaler.fit_transform(
    train_data[feature_cols]
)
val_data_scaled[feature_cols] = scaler.transform(val_data[feature_cols])
test_data_scaled[feature_cols] = scaler.transform(test_df[feature_cols])
```

Min-Max Scaling of Training, Validation, and Test Features.

```
sequence_length = 30
target_col = "RUL"

X_train, y_train = process_engines_data(
    train_data_scaled, feature_cols, sequence_length, target_col
)
X_val, y_val = process_engines_data(
    val_data_scaled, feature_cols, sequence_length, target_col
)
```

Sequence Generation for Training and Validation Data.

The scaled sensor data was converted into sequences of 30 consecutive operating cycles to capture the temporal patterns in engine behavior. Each sequence was associated with the RUL value at its final time step. The training and validation sequences were generated separately to support model training and performance monitoring.

3.2.5 BiLSTM Model Architecture

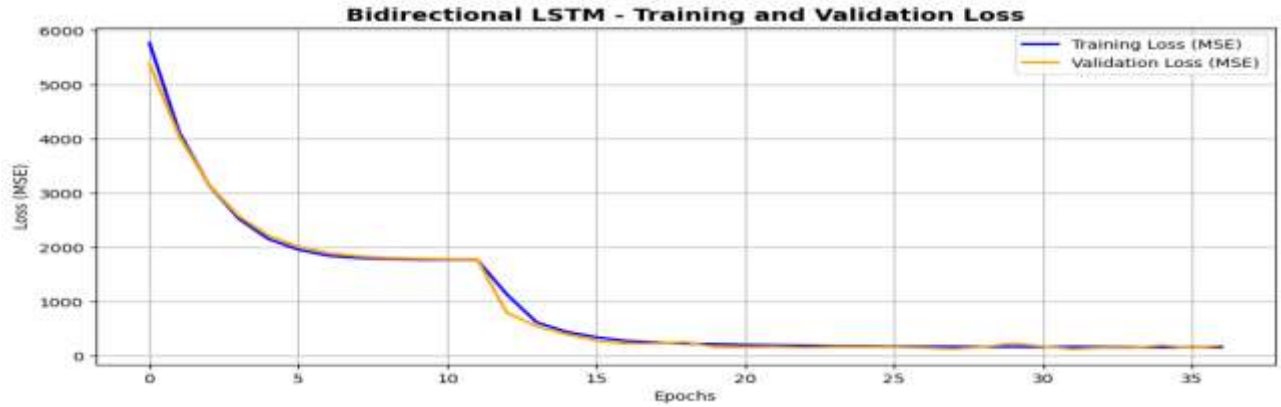
A Bidirectional Long Short-Term Memory (BiLSTM) neural network was developed to predict the Remaining Useful Life (RUL) of aircraft engines. The model was designed to learn temporal patterns from sequences of sensor measurements and use them to estimate the remaining operating cycles before failure.

The model consists of two bidirectional LSTM layers with 64 and 32 units, respectively. Dropout layers with a rate of 0.3 were added after each LSTM layer to help reduce overfitting. A final Dense layer with one linear output was used to generate the predicted RUL value.

3.2.6 Model Training

The BiLSTM model was trained using the Adam optimizer with a learning rate of 0.001 and Mean Squared Error (MSE) as the loss function. The training process was configured for a maximum of 50 epochs with a batch size of 64.

Early stopping was applied to monitor the validation loss during training. Training was stopped if the validation loss did not improve for five consecutive epochs, and the model weights from the best-performing epoch were restored. This approach helped reduce unnecessary training and limit overfitting.

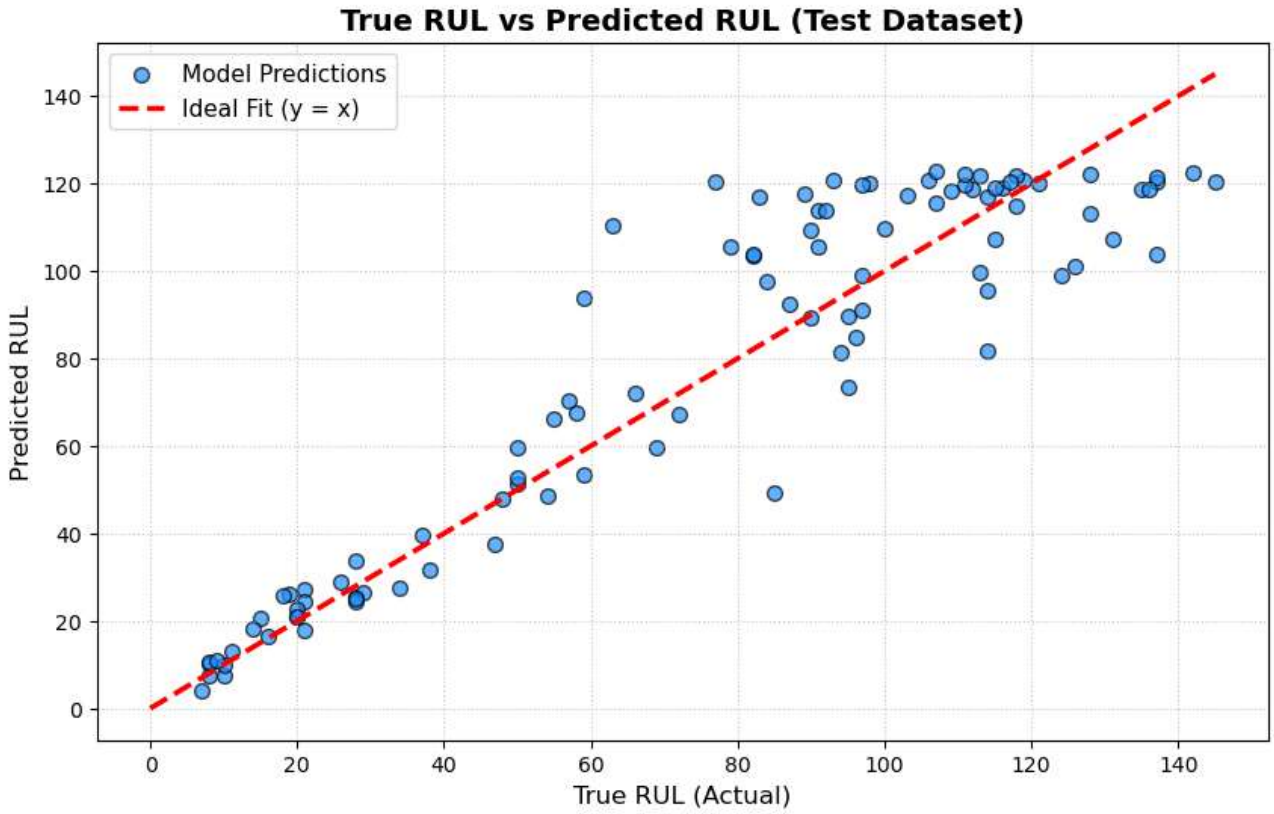


Training and Validation Loss During BiLSTM Model Training

3.2.7 Model Evaluation and Results

The Bidirectional LSTM model was evaluated on the test dataset using four performance metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), the coefficient of determination (R^2), and the NASA scoring function. The model achieved an RMSE of 15.4801 and an MAE of 11.4270. The R^2 score was 0.8612, indicating that the model explained approximately 86.12% of the variance in the test target values. The NASA scoring function returned a value of 492.39.

```
============================================================
📈 تقرير مقاييس الأداء النهائية لنموذج Bidirectional LSTM:
============================================================
• Root Mean Squared Error (RMSE) : 15.4801
• Mean Absolute Error (MAE)      : 11.4270
• R² Score (معامل التحديد)        : (%86.12)  0.8612
• NASA Official Score Function   : 492.39
============================================================
```



Actual versus Predicted Remaining Useful Life (RUL) for the FD001 Test Dataset.

3.3 FD002 Model Development and Evaluation

Project Notebook for FD002 data Set

The complete implementation of the FD002 RUL prediction model is available in the following notebook:

Notebook Link:
https://colab.research.google.com/drive/1NJ7XUPmfyUJthLjYNzDWLTsDk1dJqspO#scrollTo=63b46bef

3.3.1 Dataset Preparation and RUL Calculation

The FD002 subset of the NASA C-MAPSS dataset was used to develop a model for predicting the Remaining Useful Life (RUL) of aircraft engines. The dataset contains engine identifiers, operating cycles, three operational settings, and 21 sensor measurements.

The RUL values were calculated by subtracting each engine's current operating cycle from its maximum recorded cycle. The resulting RUL values were capped at 125 cycles to limit the maximum target value used during model development.

3.3.2 Feature Selection, Scaling, and Sequence Generation

Constant features were removed from the input feature set before applying Min-Max scaling to the remaining features. The scaled features were then organized into sequences of 50 consecutive operating cycles for each engine. Each sequence was assigned the RUL value corresponding to its final cycle.

To prepare the model inputs, the RUL target values were also scaled using Min-Max scaling. The resulting sequences were randomly divided into training and validation subsets, with 80% used for training and 20% reserved for evaluation.

3.3.3 BiLSTM Model Architecture and Training

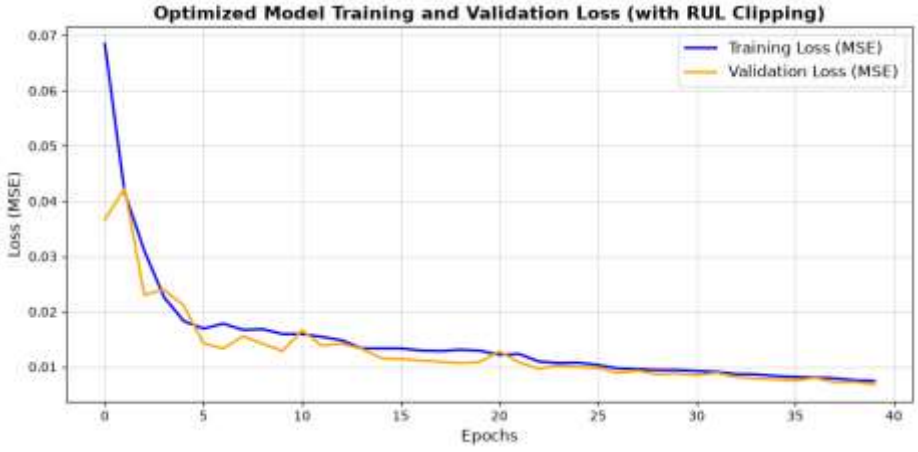
A Bidirectional Long Short-Term Memory (BiLSTM) model was developed to predict the Remaining Useful Life (RUL) of aircraft engines using sequences of sensor measurements. The model consists of two bidirectional LSTM layers with 100 and 50 units, respectively, followed by dropout layers with a rate of 0.2. A Dense layer with 50 neurons and ReLU activation was added before the final output layer, which contains one neuron with sigmoid activation.

The model was compiled using the Adam optimizer with a learning rate of 0.001 and Mean Squared Error (MSE) as the loss function. Training was configured for a maximum of 40 epochs with a batch size of 64. Early stopping was used to stop training when the validation loss did not improve for eight consecutive epochs, while ReduceLROnPlateau reduced the learning rate when the validation loss stopped improving for three consecutive epochs.

```
# 3. Model
# ----------------------------------------
bilstm_model = Sequential([
        Bidirectional(
            LSTM(100, activation="tanh", return_sequences=True), input_shape=input_shape
        ),
        Dropout(0.2),
        Bidirectional(LSTM(50, activation="tanh", return_sequences=False)),
        Dropout(0.2),
        Dense(50, activation="relu"),
        Dense(1, activation="sigmoid"),
    ])
bilstm_model.compile(optimizer=Adam(learning_rate=0.001), loss="mse", metrics=["mae"])

callbacks = [
    EarlyStopping(monitor="val_loss", patience=8, restore_best_weights=True, verbose=1),
    ReduceLROnPlateau(
        monitor="val_loss", factor=0.5, patience=3, min_lr=0.00001, verbose=1
    ),
]
```

FD002 BiLSTM Model Architecture and Training Configuration



Training and Validation Loss During FD002 Model Training

3.3.4 Model Evaluation and Results

The FD002 BiLSTM model was evaluated on the official test dataset using MAE and RMSE. The model achieved an MAE of 18.47 cycles and an RMSE of 27.83 cycles across 259 test engines.

```
...  (259, 3)
     Engine_ID  Actual_RUL  Predicted_RUL
0            1          18      12.981105
1            2          79     123.678810
2            3         106     114.517357
3            4         110     108.696304
4            5          15      16.400084
Test MAE : 18.469632932118007
Test RMSE: 27.82787792184366
```

FD002 Test Predictions and Evaluation Results

3.3.5 Prediction Export and SQL Server Integration

The predicted RUL values for the 259 FD002 test engines were exported to a CSV file named `Predicted_RUL_Results_FD002.csv`. The results were also integrated into the SQL Server database table `dbo.rul_predictions` to support centralized storage and further analysis.

3.4 FD003 Model Development and Evaluation

Project Notebook For FD003 data Set

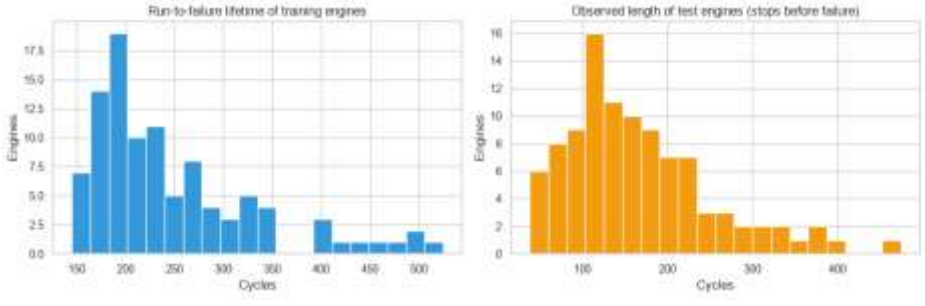
The complete implementation of the FD003 RUL prediction model is available in the following notebook:

Notebook Link:
https://colab.research.google.com/drive/1mImGl_SvhJm9xPdpPM0K_WB6uZGYm2gY#scrollTo=53be454a

3.4.1 Dataset Preparation and Exploratory Data Analysis

The FD003 subset of the NASA C-MAPSS dataset was used to develop an LSTM-based model for predicting the Remaining Useful Life (RUL) of aircraft engines. The dataset includes engine identifiers, operating cycles, three operational settings, and 21 sensor measurements.

Exploratory Data Analysis (EDA) was performed to examine the dataset structure, check for missing values, and analyze the operating lifetimes of training and test engines. Constant features were also identified to determine which input variables should be retained for model development.

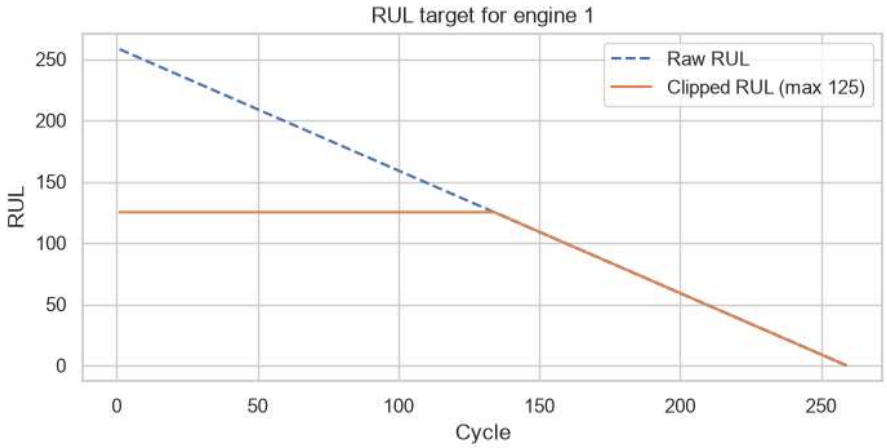


Distribution of Training and Test Engine Lifetimes in FD003

3.4.2 RUL Calculation and Feature Engineering

The Remaining Useful Life (RUL) was calculated for each training engine by subtracting its current operating cycle from its final recorded cycle. To limit the effect of very large RUL values, the calculated values were capped at 125 cycles.

Feature engineering was then performed using rolling means calculated over the previous 15 operating cycles for each engine. These rolling features were combined with the selected sensor measurements to provide the LSTM model with additional information about changes in engine behavior over time. Since FD003 represents a single operating condition, the operating settings were excluded from the model input features.



Raw and Capped RUL Values for a Training Engine in FD003

3.4.3 Data Splitting, Scaling, and Sequence Generation

The training data was divided into training and validation sets based on engine identifiers, with 80% of the engines used for training and 20% for validation. This approach ensured that the same engine did not appear in both sets.

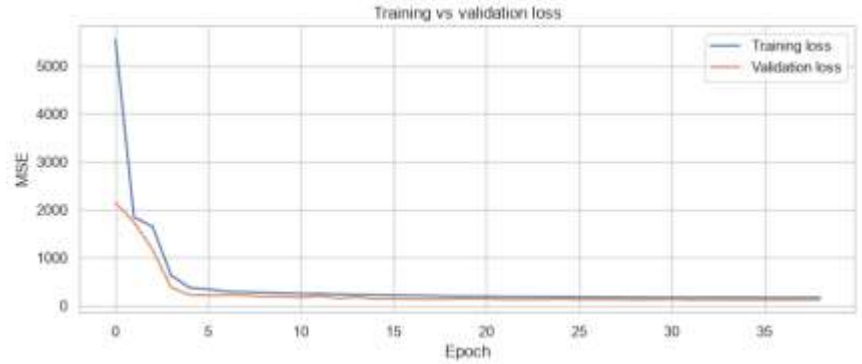
MinMaxScaler was fitted using the training data and then applied to the validation and test data to scale the input features to a consistent range. The scaled data was converted into sequences of 30 consecutive operating cycles, with each sequence associated with the RUL value at its final cycle.

For the test dataset, the last 30 available cycles of each engine were used to generate a sequence. When an engine had fewer than 30 recorded cycles, padding was applied to create a sequence of the required length.

3.4.4 LSTM Model Architecture and Training

An LSTM model was developed to predict the Remaining Useful Life (RUL) of aircraft engines. It consists of two LSTM layers with 64 and 32 units, followed by a Dense layer with 16 neurons and a final output layer.

The model was trained using the Adam optimizer, MSE loss, a batch size of 64, and a maximum of 50 epochs. Early stopping and learning rate reduction were used to improve the training process.



Training and Validation Loss During FD003 Model Training

3.4.5 Model Evaluation and Prediction Error Analysis

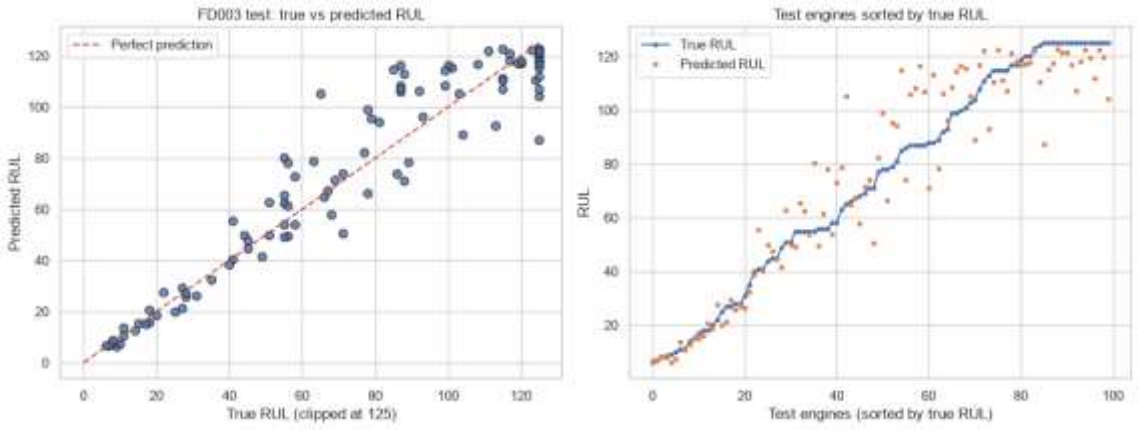
The trained LSTM model was evaluated using the official FD003 test dataset. Its performance was measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 score. The NASA scoring function was also used to evaluate prediction errors.

The predicted RUL values were compared with the actual values to assess the model's prediction accuracy and identify errors. Both capped and raw RUL values were considered during evaluation.

```
TEST vs clipped true RUL (125)     MAE   8.72 | RMSE  12.17 | R2  0.9035 | NASA     258.3
TEST vs raw true RUL               MAE  10.28 | RMSE  14.01 | R2  0.8855 | NASA     307.1
```

	unit_number	Actual_RUL_raw	Actual_RUL_clip	Predicted_RUL	Error	Abs_Error
0	1	44	44	49.910000	5.91	5.91
1	2	51	51	62.840000	11.84	11.84
2	3	27	27	29.469999	2.47	2.47
3	4	120	120	117.059998	-2.94	2.94
4	5	101	101	115.540001	14.54	14.54
5	6	99	99	108.500000	9.50	9.50
6	7	71	71	50.459999	-20.54	20.54
7	8	55	55	65.459999	10.46	10.46
8	9	55	55	62.380001	7.38	7.38
9	10	66	66	64.940002	-1.06	1.06

FD003 Test Dataset Evaluation Metrics and Predicted RUL Results



Actual vs. Predicted RUL and Prediction Error Analysis for the FD003 Test Dataset

3.4.6 Model Saving, Inference, and SQL Server Integration

The trained LSTM model, feature scaler, and configuration settings were saved for future use. A reusable prediction function was also developed to estimate RUL for individual engines using the same preprocessing steps applied during model development.

The predicted RUL results were exported to CSV files for further analysis and integration with SQL Server. The database integration code was designed to update the FD003 predictions in the `rul_predictions` table and verify the number of inserted records.

3.5 FD004 Model Development and Evaluation

Project Notebook for FD004 data Set

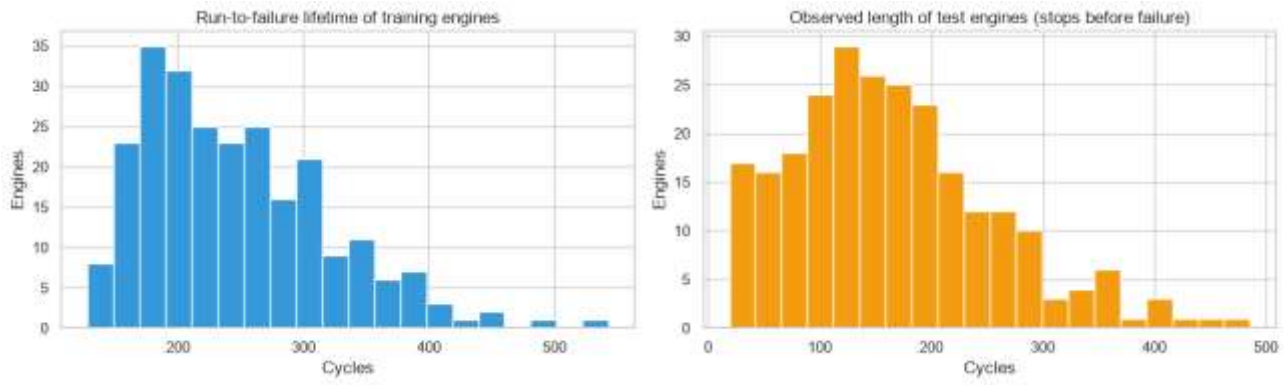
The complete implementation of the FD004 RUL prediction model is available in the following notebook:

Notebook Link:
https://colab.research.google.com/drive/1oyYCCXAHC_iEaCVQCR5_XX7rOeIYawf3#scrollTo=a50b1bba

3.5.1 Dataset Preparation and Exploratory Data Analysis

The NASA C-MAPSS FD004 dataset was used to develop an LSTM model for predicting the Remaining Useful Life (RUL) of aircraft engines. The dataset contains engine identifiers, operating cycles, three operational settings, and 21 sensor measurements.

Exploratory Data Analysis (EDA) was performed to check for missing values, examine engine lifetimes, and identify constant features. Constant features were removed before model development, and sensor measurements were selected as the model inputs.

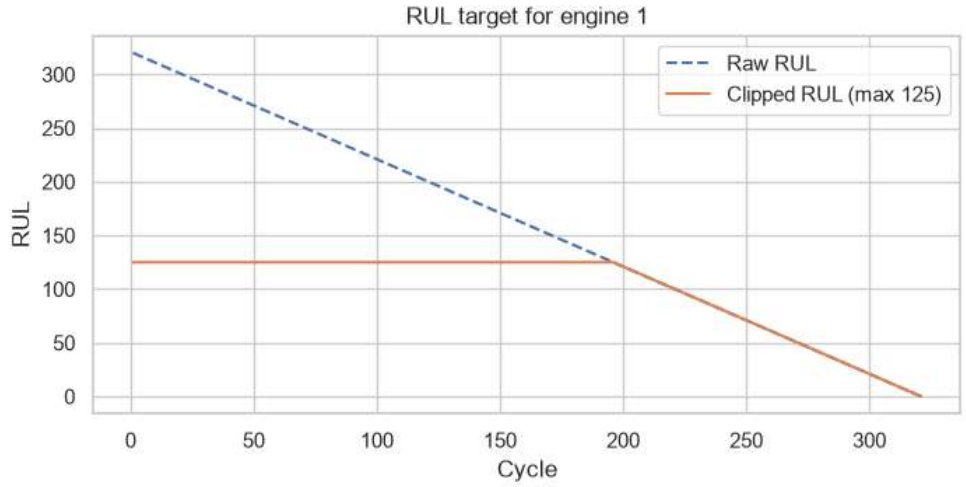


Distribution of Training and Test Engine Lifetimes in FD004

3.5.2 RUL Calculation and Feature Engineering

The Remaining Useful Life (RUL) was calculated by subtracting each engine's current cycle from its final recorded training cycle. The RUL values were capped at 125 cycles to limit large target values.

Feature engineering was performed by calculating rolling means over the previous 15 cycles for each sensor. These features were combined with the original sensor measurements to help the model capture changes in engine behavior over time.



Raw and Capped RUL Values for a Training Engine in FD004

3.5.3 Data Splitting, Scaling, and Sequence Generation

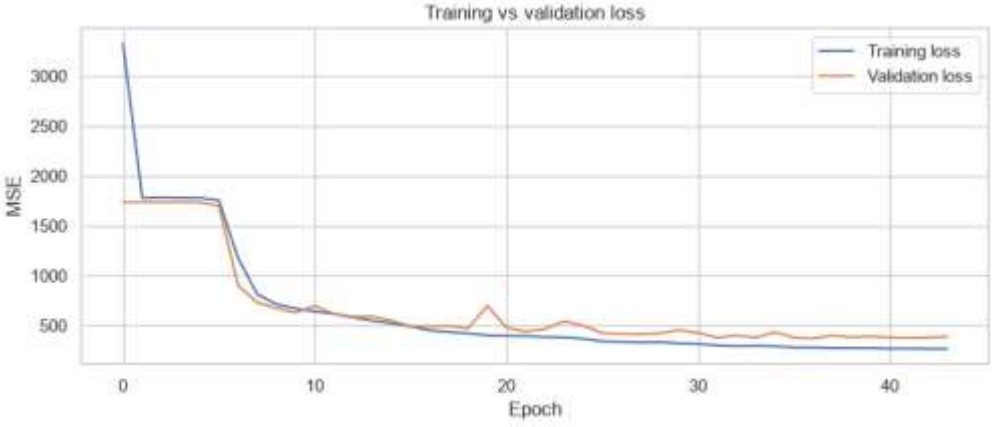
The training data was divided into training and validation sets based on engine identifiers, using 80% of the engines for training and 20% for validation. This ensured that each engine belonged to only one set.

MinMaxScaler was fitted on the training data and then applied to the validation and test data. The scaled sensor measurements were converted into sequences of 30 consecutive cycles, with each sequence representing the RUL at its final cycle. For the test dataset, the last 30 available cycles of each engine were used to generate one prediction per engine.

3.5.4 LSTM Model Architecture and Training

An LSTM model was developed to predict the Remaining Useful Life (RUL) of aircraft engines. It consists of two LSTM layers with 64 and 32 units, followed by a Dense layer with 16 neurons and a final output layer.

The model was trained using the Adam optimizer, MSE loss, a batch size of 64, and a maximum of 50 epochs. Early stopping and learning rate reduction were applied to improve the training process.



Training and Validation Loss During FD004 Model Training

3.5.5 Model Evaluation and Prediction Error Analysis

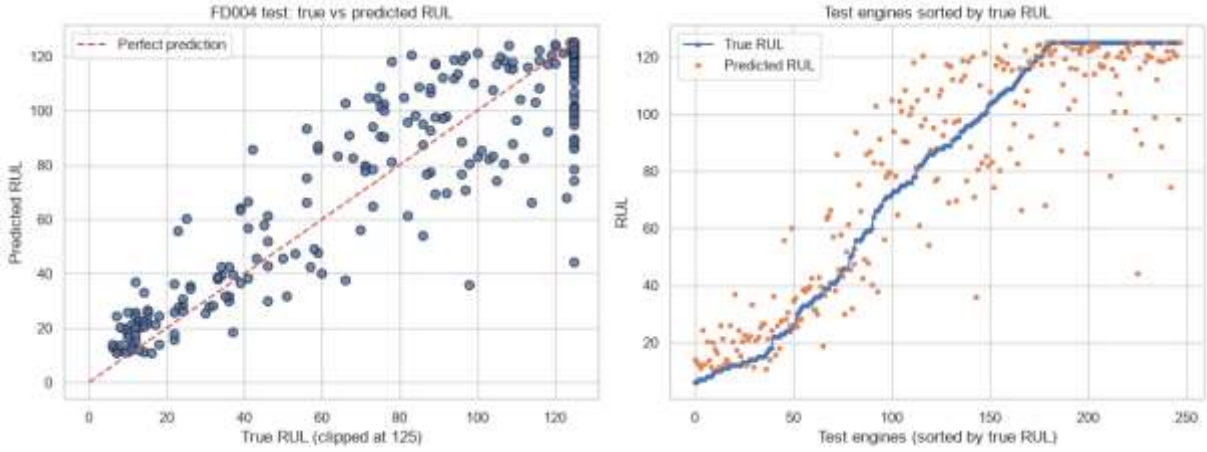
The trained LSTM model was evaluated using the official FD004 test dataset. Performance was measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), the R^2 score, and the NASA scoring function.

The predicted RUL values were compared with both raw and capped actual RUL values to assess prediction accuracy. Prediction errors were also analyzed to understand the differences between the estimated and actual engine lifetimes.

```
TEST vs clipped true RUL (125)    MAE  13.23 | RMSE  18.02 | R2  0.8242 | NASA     1842.2
TEST vs raw true RUL              MAE  21.92 | RMSE  29.17 | R2  0.7138 | NASA     9061.7
```

	unit_number	Actual_RUL_raw	Actual_RUL_clip	Predicted_RUL	Error	Abs_Error
0	1	22	22	36.259998	14.26	14.26
1	2	39	39	64.360001	25.36	25.36
2	3	107	107	118.599998	11.60	11.60
3	4	75	75	90.510002	15.51	15.51
4	5	149	125	102.029999	-22.97	22.97
5	6	78	78	118.330002	40.33	40.33
6	7	94	94	111.940002	17.94	17.94
7	8	14	14	11.320000	-2.68	2.68
8	9	99	99	110.260002	11.26	11.26
9	10	162	125	125.000000	0.00	0.00

FD004 Test Dataset Evaluation Metrics.



Actual vs. Predicted RUL for FD004 Test Engines.

3.5.6 Model Saving, Inference, and SQL Server Integration

The trained LSTM model, scaler, and configuration settings were saved for future use. A reusable function was also developed to predict the RUL of an individual engine using the same preprocessing steps as the original model.

The prediction results were exported to CSV files. SQL Server integration code was implemented to insert the FD004 predictions into the `rul_predictions` table and verify the number of stored records.

Chapter 4
Database Design, Implementation, and Power BI Dashboard

This chapter presents the design and implementation of the EngineRUL database using Microsoft SQL Server. It also describes the integration of RUL prediction results with the database and their use in the Power BI dashboard for engine health monitoring and prediction analysis.

4.1 Database Design

The project uses a relational database named `EngineRUL`, implemented in Microsoft SQL Server to store and organize the Remaining Useful Life (RUL) prediction results generated by the deep learning models.

The database consists of two tables, `dim_dataset` and `rul_predictions`, and one view, `vw_rul_dashboard`. Each component serves a specific purpose in organizing prediction results and preparing data for visualization.

The `dim_dataset` Table

The `dim_dataset` table stores descriptive information about each dataset, including its operating conditions, fault modes, and description. The `dataset` column is the primary key and uniquely identifies each dataset.

The `rul_predictions` Table

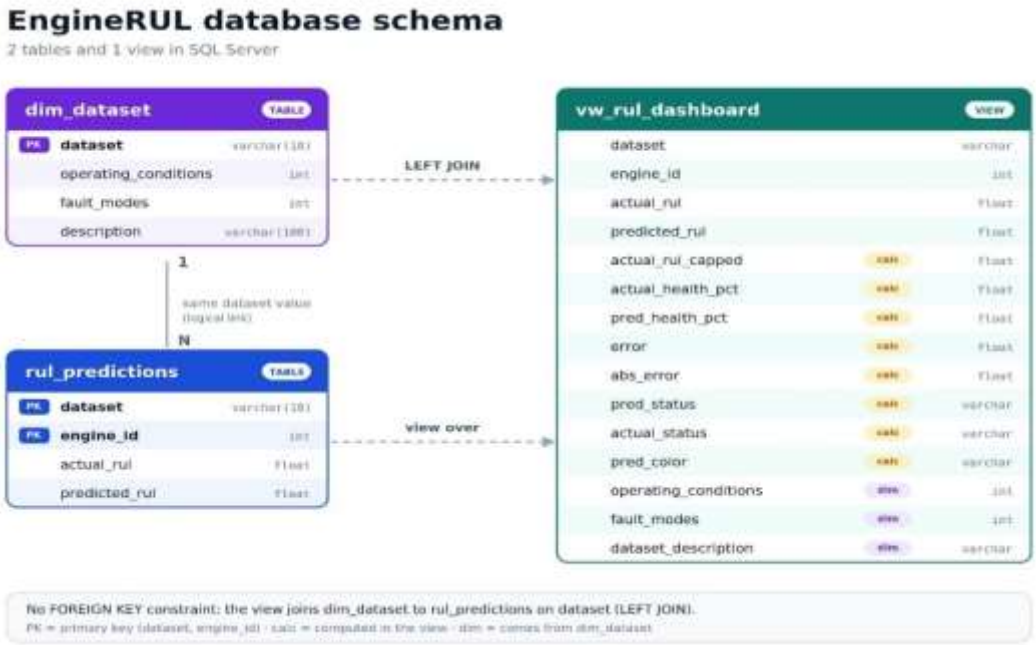
The `rul_predictions` table stores the engine-level prediction results. It contains the dataset identifier, engine ID, actual RUL, and predicted RUL. Its composite primary key consists of `dataset` and `engine_id`, ensuring that each engine has a unique record within its dataset.

The `vw_rul_dashboard` View

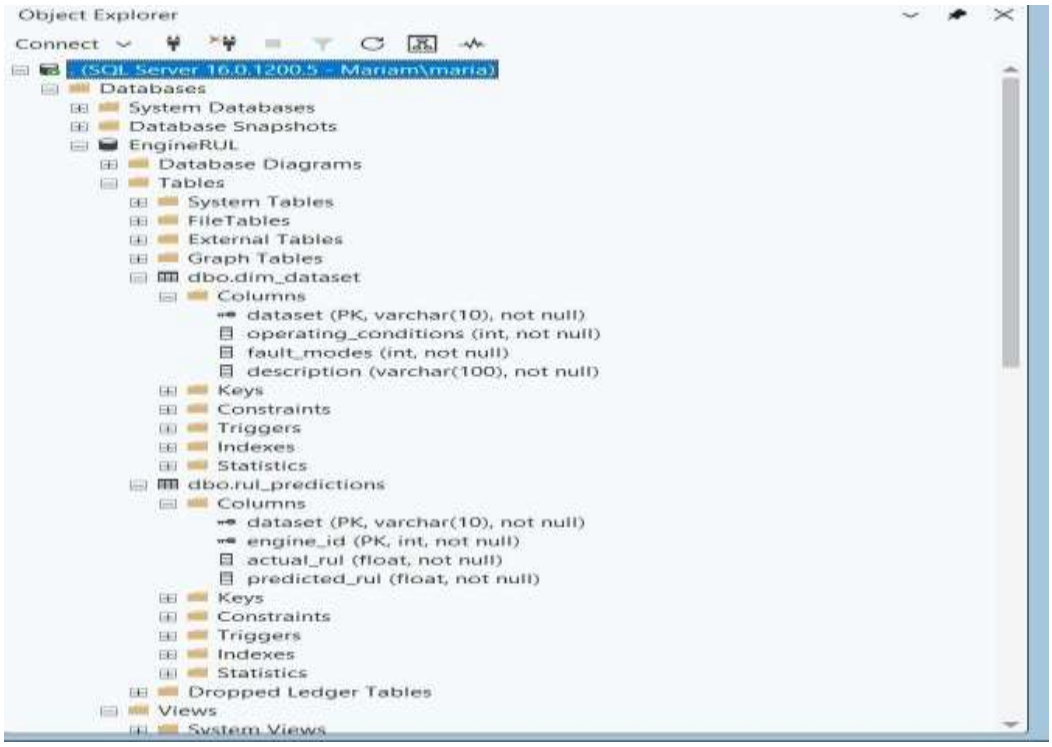
The `vw_rul_dashboard` view combines prediction records with dataset information using a `LEFT JOIN` on the `dataset` column. It also provides calculated fields, including capped actual RUL, actual and predicted health percentages, prediction error, absolute error, status classifications, and prediction color indicators.

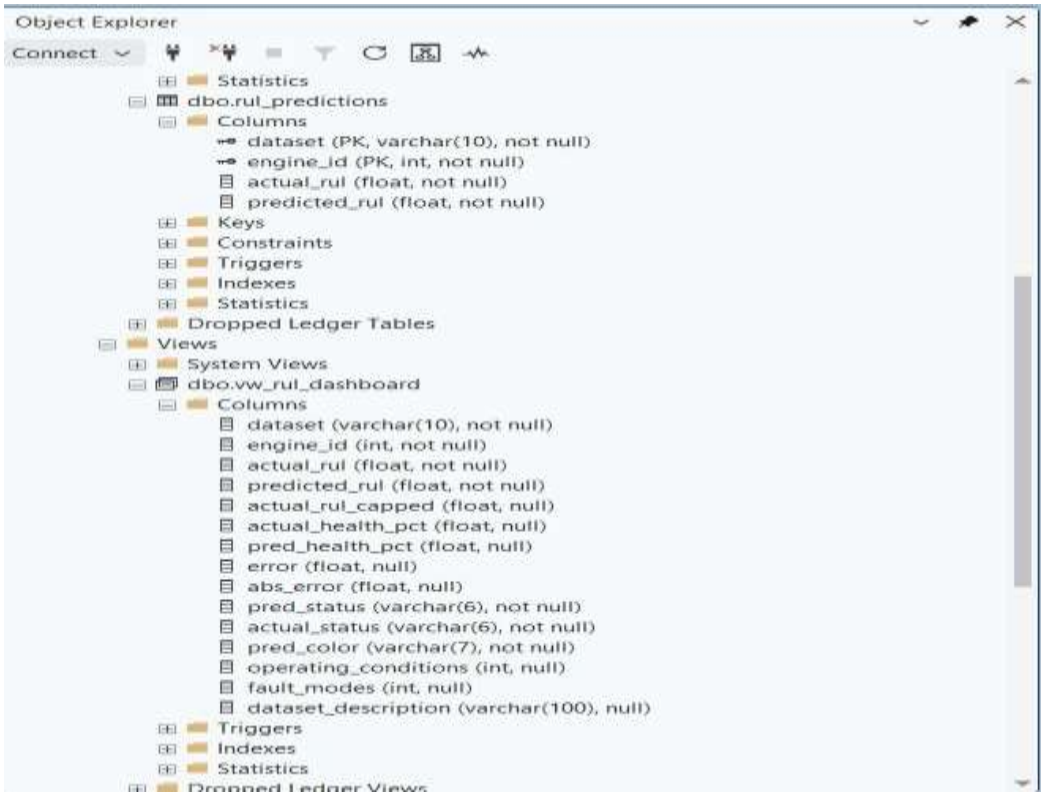
This view provides a prepared dataset for Power BI, supporting the visualization and analysis of engine health and model prediction performance.

Although the two tables are logically linked through the `dataset` column, no physical foreign key constraint is defined. The relationship is implemented within the view using the `LEFT JOIN` operation.



4.2 SQL Server Implementation





4.3 Model Predictions Integration

The prediction results generated by the deep learning models were prepared for storage in the EngineRUL database. Each record includes the dataset identifier, engine ID, actual RUL, and predicted RUL.

The results were exported to CSV files and SQL Server integration code was developed to insert the prediction records into the `rul_predictions` table. This structure allows the prediction results from different NASA C-MAPSS datasets to be organized in a single database.

The `vw_rul_dashboard` view combines the stored predictions with dataset information and provides calculated indicators, including prediction errors and engine health percentages. These fields support the analysis and visualization of model results in Power BI.

4.4 Power BI Dashboard Development

Microsoft Power BI is used to visualize the RUL prediction results stored in the `EngineRUL` SQL Server database. The dashboard uses the `vw_rul_dashboard` view to display engine information, actual and predicted RUL values, prediction errors, and engine health indicators.

The dashboard provides an interactive visual representation of the model results, supporting the analysis and monitoring of aircraft engine health.



Power BI Dashboard for Aircraft Engine RUL Prediction

4.5 Dashboard Components and Key Performance Indicators (KPIs)

The *Turbofan Health Monitor* dashboard provides an interactive overview of aircraft engine health and Remaining Useful Life (RUL) predictions across the four NASA C-MAPSS datasets: FD001, FD002, FD003, and FD004. It combines key performance indicators, engine health classifications, and visualizations to support the analysis of prediction results.

4.5.1 Interactive Filters

The dashboard includes three filters: Dataset, Engine ID, and Health Status. The Dataset filter allows users to select FD001, FD002, FD003, FD004, or all datasets. The Engine ID filter allows users to select a specific engine, while the Health Status filter enables the analysis of engines classified as Healthy, Warning, or Critical.

4.5.2 Header KPIs

The dashboard displays four key indicators: Actual Remaining Cycles, Predicted Remaining Cycles, Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). The displayed

average actual RUL is 75.27 cycles, while the average predicted RUL is 75.51 cycles. The reported RMSE is 16.31, and the MAE is 11.37 cycles.

The difference between the average actual and predicted RUL is 0.24 cycles. Although these averages are close, they do not fully represent the accuracy of individual predictions. MAE measures the average absolute prediction error, while RMSE gives greater weight to larger errors.

4.5.3 Fleet Health KPIs

The dashboard summarizes the health status of 707 engines across the four datasets. It displays 437 Healthy engines (61.8%), 146 Warning engines (20.6%), and 124 Critical engines (17.5%).

These indicators provide an overview of the fleet's health-status distribution. The classification of each engine depends on the RUL thresholds configured in the dashboard and is intended to support monitoring and analysis.

4.5.4 Actual vs. Predicted RUL

The Actual vs. Predicted RUL scatter plot compares actual RUL values on the horizontal axis with predicted RUL values on the vertical axis. A dashed diagonal line represents perfect prediction, where predicted RUL equals actual RUL. Points closer to this line indicate smaller prediction errors.

Different colors represent Critical, Warning, and Healthy engines. The plot helps identify patterns in prediction performance and individual engines with relatively large errors. A concentration of points at an actual RUL of 125 is consistent with RUL clipping at 125, although the preprocessing and evaluation procedures should be checked to confirm this interpretation.

4.5.5 Fleet Risk Distribution

The Fleet Risk Distribution visual is a 100% stacked bar chart that compares the proportions of Healthy, Warning, and Critical engines across the four datasets.

FD001 contains 64 Healthy, 20 Warning, and 16 Critical engines. FD002 contains 151 Healthy, 52 Warning, and 56 Critical engines. FD003 contains 62 Healthy, 22 Warning, and 16 Critical engines. FD004 contains 160 Healthy, 52 Warning, and 36 Critical engines.

The chart shows that Healthy engines represent the largest group in each dataset. The status counts sum to 707 engines, matching the totals displayed in the Fleet Health KPI cards.

4.5.6 Model Error by Dataset

The Model Error by Dataset visual compares MAE and RMSE for each dataset. The displayed values are summarized below.

Dataset	MAE	RMSE
FD001	10.5	14.8
FD002	11.0	16.5
FD003	8.7	12.2
FD004	13.2	18.0

RMSE is higher than MAE for all four datasets, reflecting its greater sensitivity to large prediction errors. These values summarize the errors reported for each dataset and should be interpreted in the context of the corresponding model evaluation procedures.

Overall, the dashboard integrates RUL prediction summaries, engine health indicators, and model error visualizations into a single interactive interface. It supports the exploration of engine health and prediction performance across the four NASA C-MAPSS datasets.

Chapter 5
Conclusion and Future Work

This chapter introduces the project, defines the problem statement and objectives, and outlines the project scope and proposed approach to aircraft engine predictive maintenance using deep learning.

5.1 Conclusion

This project developed a predictive maintenance system for aircraft engines using the NASA C-MAPSS datasets. Deep learning models based on LSTM and BiLSTM architectures were trained to predict the Remaining Useful Life (RUL) of aircraft engines across four dataset subsets: FD001, FD002, FD003, and FD004.

The model predictions were evaluated using MAE, RMSE, R^2, and NASA Score. SQL Server was used to store and organize prediction results, while Power BI provided visualizations to support the analysis of model performance and engine health indicators.

Overall, the project demonstrates how deep learning, database management, and business intelligence tools can be integrated to support data-driven aircraft engine maintenance planning.

5.2 Future Work

Several practical features could be added to extend the capabilities of the proposed aircraft engine predictive maintenance system.

1. Sensor Fault Identification and Explainability

The system could be extended to identify which sensors contribute most to a low RUL prediction or an abnormal engine condition. By analyzing sensor trends and their influence on model predictions, the system could help maintenance engineers investigate potential causes of engine deterioration. This feature would provide additional insight beyond predicting the Remaining Useful Life alone. However, identifying influential sensors would not necessarily confirm the actual physical cause of a fault.

2. LLM-Based Maintenance Chatbot

A Large Language Model (LLM)-based chatbot could be integrated into the system to allow users to ask natural-language questions about engine health and prediction results. For example, users could ask about a specific engine's predicted RUL, its health status, or the sensors associated with an abnormal prediction. The chatbot could retrieve relevant information from the SQL Server database and explain the available results in a user-friendly format.

3. Sensor Anomaly Detection

An anomaly detection component could be developed to identify unusual sensor readings or changes in sensor behavior. This feature could help highlight engines whose sensor measurements differ significantly from expected operating patterns, even before a low RUL prediction is produced.

4. Maintenance Recommendation System

The system could be extended to generate maintenance recommendations based on predicted RUL, engine health status, and sensor analysis. For example, engines with critically low predicted RUL could be prioritized for inspection. Recommendations would require appropriate engineering rules and validation before being used in real maintenance decisions.

5. Automated Alerts and Notifications

An alerting mechanism could notify users when an engine's predicted RUL falls below a predefined threshold or when significant sensor anomalies are detected. Alerts could be displayed on the

dashboard or delivered through an appropriate notification service, helping users identify engines that require further investigation.

6. Real-Time Engine Monitoring

Future development could connect the system to continuously or periodically collected engine sensor data. The models could then generate updated RUL predictions, store the results in SQL Server, and refresh the dashboard automatically. This would move the project toward a more dynamic predictive maintenance platform.

These extensions could transform the current prediction and visualization workflow into a more comprehensive decision-support system by combining RUL estimation, sensor analysis, natural-language interaction, and maintenance planning.